

Cambridge (CIE) IGCSE Additional Maths



Your notes

Arithmetic & Geometric Progressions

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Your notes

Language of Sequences & Series

Language of Sequences & Series

What is a progression?

- A **progression** (also called a **sequence**) is an ordered set of numbers with a rule for finding all of the numbers in the sequence
 - For example 1, 3, 5, 7, 9, ... is a sequence with the rule 'start at one and add two to each number'
- The numbers in a progression are often called **terms**
- The terms of a progression are often referred to by letters with a subscript
 - This will often be the letter u
 - So in the progression above, $u_1 = 1$, $u_2 = 3$, $u_3 = 5$ and so on
- Each term in a progression can be found by **substituting** the term number into **formula for the n^{th} term**

What is a series?

- You get a **series** by summing up the terms in a progression
 - E.g. For the sequence 1, 3, 5, 7, ... the associated series is $1 + 3 + 5 + 7 + \dots$
- We use the notation S_n to refer to the sum of the first n terms in the series
 - $S_n = u_1 + u_2 + u_3 + \dots + u_n$
 - So for the series above $S_5 = 1 + 3 + 5 + 7 + 9 = 25$



Worked Example

Determine the first five terms and the value of S_5 in the progression with terms defined by $u_n = 5 - 2n$.

Substitute n in for each term that you want to find.



$$u_1 = 5 - 2(1) = 3$$

$$u_2 = 5 - 2(2) = 1$$

$$u_3 = 5 - 2(3) = -1$$

$$u_4 = 5 - 2(4) = -3$$

$$u_5 = 5 - 2(5) = -5$$

The first five terms are 3, 1, -1, -3, -5

To find S_5 add the first five terms of the progression together.

$$3 + 1 + (-1) + (-3) + (-5)$$

$$S_5 = -5$$

Sigma Notation

What is sigma notation?

- Sigma notation is used to show the sum of a certain number of terms in a sequence
- The symbol Σ is the capital Greek letter sigma
- Σ stands for 'sum'
- The expression to the right of the Σ tells you what is being summed, and the limits above and below tell you which terms you are summing

THESE LIMITS TELL YOU THAT YOU ARE SUMMING $(2r-1)$ USING THE VALUES $r=1, r=2, \dots$ UP TO $r=5$

$$\sum_{r=1}^5 (2r-1) = 1 + 3 + 5 + 7 + 9$$

SUBSTITUTE $r=1, r=2, r=3, r=4, r=5$ INTO $(2r-1)$ TO FIND THE FIVE TERMS THAT ARE BEING SUMMED

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- Be careful, the limits don't have to start with 1

- For example $\sum_{k=0}^4 (2k+1)$ or $\sum_{k=7}^{14} (2k-13)$

- r and k are commonly used variables within sigma notation



Examiner Tips and Tricks

- Sigma notation will not be tested in the exam but understanding it will help you to further understand series



Your notes



Your notes

Arithmetic Progressions

Arithmetic Sequences

What is an arithmetic progression?

- In an **arithmetic progression** (also called **arithmetic sequence**), the difference between consecutive terms in the sequence is constant
- This **constant difference** is known as the **common difference**, d , of the sequence
- For example, 1, 4, 7, 10, ... is an arithmetic sequence with the rule 'start at one and add three to each number'
 - The **first term**, a , is 1
 - The **common difference**, d , is 3
- An arithmetic progression can be **increasing** (positive common difference) or **decreasing** (negative common difference)
- Each term of an arithmetic progression is referred to by the letter u with a subscript determining its place in the sequence

EXAMPLES OF ARITHMETIC SEQUENCES

$$\begin{array}{ccccccc} 2, & 5, & 8, & 11, & 14, & \dots \\ & \swarrow & \downarrow & \swarrow & \downarrow & \\ & +3 & +3 & +3 & +3 & \end{array}$$

FIRST TERM IS 2, COMMON DIFFERENCE IS 3

$$\begin{array}{ccccccc} 2, & 0.5, & -1, & -2.5, & -4, & \dots \\ & \swarrow & \downarrow & \swarrow & \downarrow & \\ & -1.5 & -1.5 & -1.5 & -1.5 & \end{array}$$

FIRST TERM IS 2, COMMON DIFFERENCE IS -1.5

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How do I find a term in an arithmetic progression?

- The **n^{th} term formula** for an arithmetic progression is given as

$$u_n = a + (n - 1)d$$

- Where **a** is the first term, and **d** is the common difference
 - This is given on the list of formulas page of the exam, you do not need to know how to derive it
- Sometimes you will be given a term and asked to find the first term or the common difference
 - Substitute the information into the formula and solve the equation
- Sometimes you will be given two terms and asked to find both the first term and the common difference
 - Substitute the information into the formula and set up two **simultaneous** equations
 - Solve the simultaneous equations



Examiner Tips and Tricks

- Simultaneous equations are often needed within arithmetic progression questions
 - Make sure you are confident solving them



Worked Example



Your notes



Your notes



The first three terms in an arithmetic sequence are $12, k^2, 5k, \dots$

Find two possible values of k .

THIS IS AN ARITHMETIC SEQUENCE SO THERE'S A SINGLE COMMON DIFFERENCE

$$d = k^2 - 12$$

AND

$$d = 5k - k^2$$

2nd TERM MINUS 1st TERM

3rd TERM MINUS 2nd TERM

SO $k^2 - 12 = 5k - k^2$

$$2k^2 - 5k - 12 = 0$$

$$(2k+3)(k-4) = 0$$

SOLVE THE QUADRATIC

$$k = -\frac{3}{2} \quad \text{OR} \quad k = 4$$

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Arithmetic Series

What is an arithmetic series?

- An arithmetic series is the **sum** of the terms in an arithmetic progression
 - It is often referred to as the **sum of an arithmetic progression**
 - For the arithmetic sequence $1, 4, 7, 10, \dots$ the arithmetic series is $1 + 4 + 7 + 10 + \dots$

How do I find the sum of an arithmetic progression?

- Use the following formulae to find the sum of the first n terms of the arithmetic series:

$$S_n = \frac{1}{2}n\{a + l\} = \frac{1}{2}n\{2a + (n-1)d\}$$



Your notes

- a is the first term
 - l is the last term
 - d is the common difference
 - n is the number of terms in the series
 - Both formulae are given on the formula page, you do not need to know how to derive them
- You can use whichever formula is more convenient for a given question
 - If you know the **first term** and **common difference** use the second version
 - If you know the **first and last term** then the first version is easier to use
- A question will often give you the sum of a certain number of terms and ask you to find the value of the first term or the common difference
 - Substitute the information into the formula and solve the equation



Examiner Tips and Tricks

- The arithmetic series formulae are given on page 2 of the exam paper – you don't need to memorise them
 - Practise finding the formulae so that you can quickly locate them in the exam



Worked Example



Your notes



The ninth term of an arithmetic progression is 28. The fifteenth term is 46.

The sum of the first n terms is 2500.

a) Show that $3n^2 + 5n - 5000 = 0$.

b) Hence find the value of n .

a) $a + (9 - 1)d = a + 8d = 28$ ①
 $a + (15 - 1)d = a + 14d = 46$ ②

SIMULTANEOUS EQUATIONS

$6d = 18$
 $d = 3$

② MINUS ①

$a = 28 - 8(3)$
 $a = 4$

SUBSTITUTE INTO ①

SO $\frac{n}{2} (2(4) + (n - 1)(3)) = 2500$
 $n(3n + 5) = 5000$
 $3n^2 + 5n - 5000 = 0$

S_n FORMULA

b) $(3n + 125)(n - 40) = 0$
 $n = -\frac{125}{3}$ OR $n = 40$

SOLVE THE QUADRATIC – YOU COULD ALSO USE THE QUADRATIC FORMULA

BUT n MUST BE A POSITIVE INTEGER

$n = 40$

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Your notes

Geometric Progressions

Geometric Sequences

What is a geometric progression?

- In a **geometric progression** (also called **geometric sequence**) there is a **common ratio**, r , between consecutive terms in the sequence
 - For example, 2, 6, 18, 54, 162, ... is a progression with the rule 'start at two and multiply each number by three'
 - The **first term**, a , is 2
 - The **common ratio**, r , is 3
- A geometric progression can be **increasing** ($r > 1$) or **decreasing** ($0 < r < 1$)
- If the common ratio is a **negative number** the terms will alternate between positive and negative values
 - For example, 1, -4, 16, -64, 256, ... is a sequence with the rule 'start at one and multiply each number by negative four'
 - The **first term**, a , is 1
 - The **common ratio**, r , is -4
- Each term of a geometric progression is referred to by the letter u with a subscript determining its place in the sequence

How do I find a term in a geometric progression?

- The n^{th} **term formula** for a geometric progression is given as

$$u_n = ar^{n-1}$$

- Where a is the first term, and r is the common ratio
 - This formula allows you to find **any term** in the geometric progression
 - It is given in the list of formulas, you do not need to know how to derive it
- Enter the information you have into the formula and find the value of the term
- Sometimes you will be given a term and asked to find the first term or the common ratio

- Substitute the information into the formula and solve the equation
- Sometimes you will be given two or more consecutive terms and asked to find both the first term and the common ratio
 - Find the common ratio by dividing a term by the one before it
 - Substitute this and one of the terms into the formula to find the first term
- Sometimes you may be given a term and the formula for the n^{th} term and asked to find the value of n
 - You can solve these using **logarithms**



Examiner Tips and Tricks

- The formula is given in the list of formulas
- If you know two terms in a geometric progression you can find **a** and **r** using simultaneous equations



Worked Example



Your notes



Your notes



The first three terms in a geometric sequence are

$2 - x$, $3x$, and x^2 respectively where $x < 0$.

a) Show that $x^3 + 7x^2 = 0$.

b) Find the 6th term of the sequence.

a) THIS IS A GEOMETRIC SEQUENCE SO
THERE'S A SINGLE COMMON RATIO

$$r = \frac{3x}{2-x}$$

2nd TERM DIVIDED BY 1st TERM

AND

$$r = \frac{x^2}{3x}$$

3rd TERM DIVIDED BY 2nd TERM

$$\text{SO } \frac{3x}{2-x} = \frac{x^2}{3x}$$

$$(3x)^2 = x^2(2-x)$$

$$9x^2 = 2x^2 - x^3$$

$$x^3 + 7x^2 = 0$$

SOLVE THE EQUATION
FROM a)

b) $x^2(x+7) = 0$

$$x = 0 \text{ OR } x = -7$$

$$\text{BUT } x < 0 \text{ SO } x = -7$$

$$a = 2 - (-7) = 9$$

FIND a AND r

$$r = \frac{3(-7)}{2-(-7)} = \frac{-21}{9} = \frac{-7}{3}$$

$$u_6 = 9 \left(\frac{-7}{3} \right)^5 = -\frac{16807}{27}$$

$u_n = ar^{n-1}$

$$\text{THE SIXTH TERM IS } -\frac{16807}{27}$$



Your notes

Geometric Series

What is a geometric series?

- A geometric series is the **sum** of the terms in a geometric progression
 - It is often referred to as the **sum of a geometric progression**
 - For the arithmetic sequence 1, 4, 7, 10, ... the arithmetic series is $1 + 4 + 7 + 10 + \dots$

How do I find the sum of a geometric progression?

- The following formulae will let you find the sum of the first n terms of a geometric progression:

$$S_n = \frac{a(1 - r^n)}{1 - r} \quad \text{or} \quad S_n = \frac{a(r^n - 1)}{r - 1}$$

- a is the first term
 - r is the common ratio ($r \neq 1$)
- The **first is given** in the list of formulas and is more convenient if $r < 1$
 - the second is more convenient if $r > 1$
- A question will often give you the sum of a certain number of terms and ask you to find the value of the first term, the common ratio, or the number of terms within the sequence
 - Substitute the information into the formula and solve the equation

What is the sum to infinity of a geometric series?

- If $|r| < 1$, then the sum of a geometric progression **converges** to a finite value given by the formula

$$S_\infty = \frac{a}{1 - r}$$

- S_∞ is known as the **sum to infinity**
- If $|r| \geq 1$ the sum of a geometric progression is **divergent** and the sum to infinity does not exist



Examiner Tips and Tricks

- The geometric series formulae are in the list of formulas on page 2 – you don't need to memorise them
 - Make sure you can locate them quickly before going into the exam



Worked Example



Your notes



Your notes



A geometric progression has first term a .

Each term is multiplied by 0.9 to get the next term.

The sum of the first 5 terms is 409.51.

- Find the value of a .
- What is the sum to infinity of the progression?

a) IDENTIFY KEY INFORMATION

$$a = a$$

$$r = 0.9$$

$$n = 5$$

$$S_5 = 409.51$$

SUBSTITUTE INTO THE FORMULA $S_n = \frac{a(1-r^n)}{1-r}$

$$409.51 = \frac{a(1-0.9^5)}{1-0.9}$$

EVALUATE USING CALCULATOR

$$409.51 = 4.0951a$$

$$a = \frac{409.51}{4.0951}$$

$$a = 100$$

b) USE FORMULA $S_\infty = \frac{a}{1-r}$

$$S_\infty = \frac{100}{1-0.9}$$

$$S_\infty = 1000$$

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